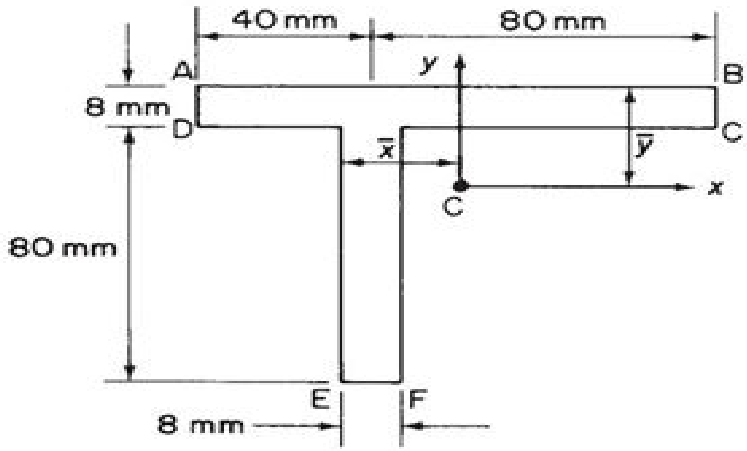
Tutorial 1: Unsymmetric bending by 26AE91R03

Problem 1:

1. A beam having the cross-section shown in Figure is subjected to a bending moment of 1500 Nm in a vertical plane. Calculate the maximum direct stress due to bending stating point at which it acts.




```
%% Finding out the CG
% Area of each of the section in mm2
A1= 120*8; A2 = 8*80;
x_cg = (A1*60+A2*40)/(A1+A2);  % from left
y_cg = (A1*4+A2*48)/(A1+A2);  % from top
x_bar= x_cg-(40-4); y_bar=y_cg;  % according to the picture
% floating-point with 2 decimals
fprintf('x_bar is %.4f and y_bar is %.4f\n', x_bar, y_bar);
```

```
x_bar is 16.0000 and y_bar is 21.6000
```

```
%%Finding Is in mm4
Ixx = (120*(8^3))/12 + A1*(4-y_cg)^2 + (8*80^3)/12 + A2*(48-y_cg)^2
```

```
Ixx =
1.0899e+06
```

```
Iyy = (8*120^3)/12 + A1*(60-x_cg)^2 + (80*8^3)/12 + A2*(40-x_cg)^2  % Calculate Iyy
```

```
Iyy =
1.3090e+06
```

```
Ixy = A1*(x_cg-60)*(-(y_cg-4)) + A2*(x_cg-40)*(48-y_cg)
```

```
Ixy =
337920
```

```
% As Only Vertical plane moment is acting so
Mx = 1500000; My=0;
% Calculate the stress components
```

```
x_coeff = (My*Ixx - Mx*Ixy)/(Ixx*Iyy-Ixy^2)
```

```
x_coeff =
-0.3862
```

```
y_coeff = (Mx*Iyy - My*Ixy)/(Ixx*Iyy-Ixy^2)
```

```
y_coeff =
1.4960
```

```
% Calculating the angle of the of nutral axis
alpha = atan((My*Ixx - Mx*Ixy)/(Mx*Iyy - My*Ixy))*180/pi % in degrees
```

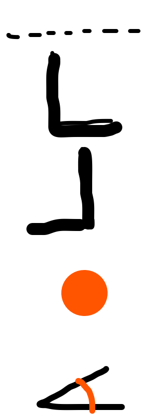
```
alpha =
-14.4748
```

```
beta = 90-alpha
```

```
beta =
104.4748
```

```
% making a function for the stress function

sigma_z = @(x,y)(x_coeff*x + y_coeff*y);

[X, Y] = meshgrid(-60:0.5:70, -70:0.5:25);

% Masks for T-section geometry
inFlange = (X >= -x_cg) & (X <= 120-x_cg) & (Y >= y_cg-8) & (Y <= 21.6);
inWeb    = (X >= 36-x_cg) & (X <= 44-x_cg) & (Y >= -88+y_cg) & (Y <= y_cg-8);
inSection = inFlange | inWeb;

Z = sigma_z(X, Y);
Z(~inSection) = NaN; % Hide points outside the section

% Plotting
figure;
contourf(X, Y, Z, 25, 'LineColor', 'none');
hold on;
colormap(jet); colorbar;
title('\sigma_z Bending Stress Distribution (MPa)');
xlabel('x (mm)'); ylabel('y (mm)');

% CG and Limited Axes
plot(0, 0, 'ro', 'MarkerSize', 10, 'MarkerFaceColor', 'r', 'DisplayName', 'CG');
plot([0, 0], [0, 20], 'k-', 'LineWidth', 2, 'DisplayName', 'Vertical Axis (x = 0)');
plot([0, 20], [0, 0], 'k-', 'LineWidth', 2, 'DisplayName', 'Horizontal Axis (y =
0)');

% Plot Neutral Axis
x_na = -60:1:60;
y_na = -tan(alpha * pi / 180) * x_na;
plot(x_na, y_na, 'k--', 'LineWidth', 2, 'DisplayName', 'Neutral Axis');
```

```matlab
% ==========================================================================
% NEW: ADD ANGLE ARC BETWEEN Y-AXIS AND NEUTRAL AXIS
% ==========================================================================

% 1. Angles in polar coordinates (radians)
% Vertical axis (x = 0, y > 0) is at 90 degrees (pi/2)
theta_yAxis = pi / 2;

% Angle of the upper/positive-y segment of the Neutral Axis
theta_NA = atan2(-tan(alpha * pi / 180) * 1, 1);
if theta_NA < 0
    theta_NA = theta_NA + pi; % Ensure we take the branch in the upper half-plane
end

% 2. Generate arc points between the y-axis and Neutral Axis
r_arc = 12; % Radius of the arc (adjust size as needed)
t_arc = linspace(min(theta_yAxis, theta_NA), max(theta_yAxis, theta_NA), 100);
arc_x = r_arc * cos(t_arc);
arc_y = r_arc * sin(t_arc);

% Plot the arc line
plot(arc_x, arc_y, 'r-', 'LineWidth', 1.5, 'DisplayName', 'Angle \alpha');

% 3. Add text label at the midpoint of the arc
mid_theta = (theta_yAxis + theta_NA) / 2;
label_x = (r_arc + 3) * cos(mid_theta);
label_y = (r_arc + 3) * sin(mid_theta);

text(label_x, label_y, sprintf('\\beta = %.2f°', abs(90+alpha)), ...
    'FontSize', 11, 'FontWeight', 'bold', 'Color', 'k', ...
    'HorizontalAlignment', 'center');

% ==========================================================================

axis equal;
grid on;
```



Neutral Axis

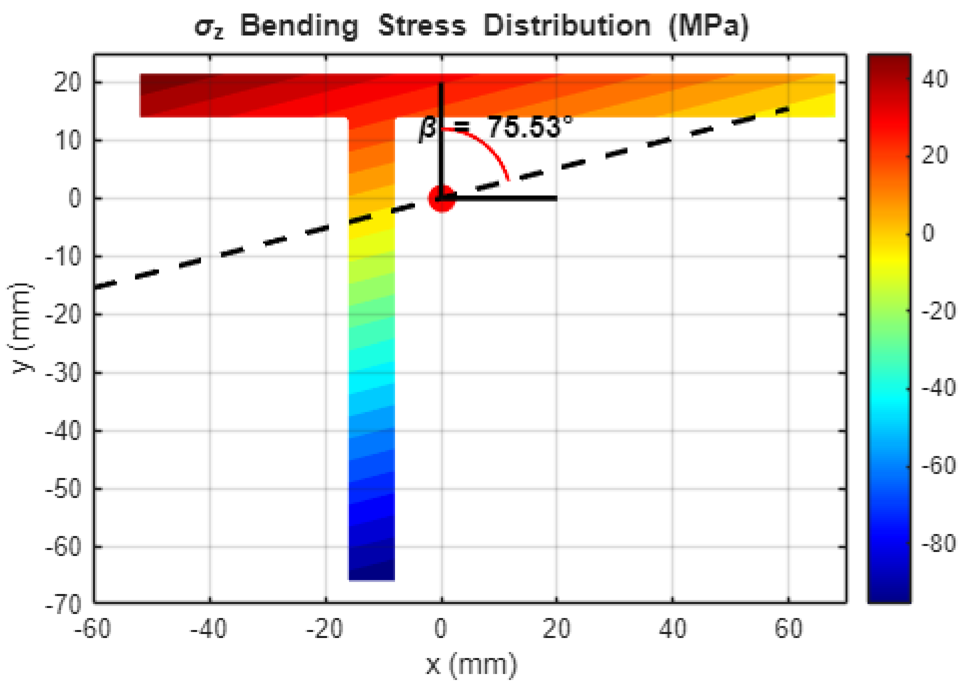
x-y coord. axis

y-z coord. axis

centroid of cross section

angle made from vertical axis (y)

Legends




```
%legend('show');
sigma_z_E = sigma_z(36-x_cg,y_cg-88)
```

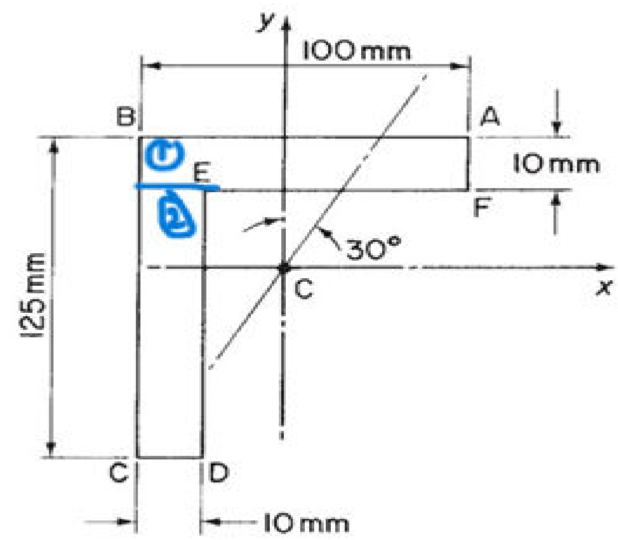
```
sigma_z_E =
-93.1581
```

```
sigma_z_F = sigma_z(44-x_cg,y_cg-88) % max
```

```
sigma_z_F =
-96.2478
```

Prob 2:

2. Figure shows the section of an angle purlin. A bending moment of 3000 N m is applied to the purlin in a plane at an angle of 30° to the vertical y axis. If the sense of the bending moment is such that its components Mx and My both produce tension in the positive xy quadrant, calculate the maximum direct stress in the purlin clearly stating the point at which it acts.




```
%% Finding out the CG
% Area of each of the section in mm2
A1= 10*100; A2 = 115*10;
x_cg = (A1*50+A2*5)/(A1+A2);  % from left
y_cg = (A1*5+A2*(0.5*115 +10))/(A1+A2);  % from top
x_bar= x_cg; y_bar=y_cg;  % according to the picture ( FROM LEFT AND TOP)
% floating-point with 2 decimals
fprintf('x_bar is %.4f and y_bar is %.4f\n', x_bar, y_bar);
```

```
x_bar is 25.9302 and y_bar is 38.4302
```

```
%%Finding Is in mm4
Ixx = (100*(10^3))/12 + A1*(5-y_cg)^2 + (10*115^3)/12 + A2*(115*0.5+10-y_cg)^2
```

```
Ixx =
3.3651e+06
```

```
Iyy = (10*100^3)/12 + A1*(50-x_cg)^2 + (115*10^3)/12 + A2*(5-x_cg)^2  % Calculate
Iyy
```

```
Iyy =
1.9261e+06
```

```
Ixy = A1*(x_cg-50)*(-y_cg+5) + A2*(x_cg-5)*(115*0.5+10-y_cg)
```

```
Ixy =
1.5044e+06
```

```
% Moment acting on the plane 30 to the vertical plane (30 CW)
M = 3000*1000  ; % N mm at the
Mx = M*cosd(30); My = M*sind(30);  % -ve sign is considered in the rest part
```

```
% Calculate the stress components
x_coeff = (My*Ixx - Mx*Ixy)/(Ixx*Iyy-Ixy^2)
```

```
x_coeff =
0.2701
```

```
y_coeff = (Mx*Iyy - My*Ixy)/(Ixx*Iyy-Ixy^2)
```

```
y_coeff =
0.6513
```

```
% Calculating the angle of the of nutral axis
alpha = atan((My*Ixx - Mx*Ixy)/(Mx*Iyy - My*Ixy))*180/pi % in degrees
```

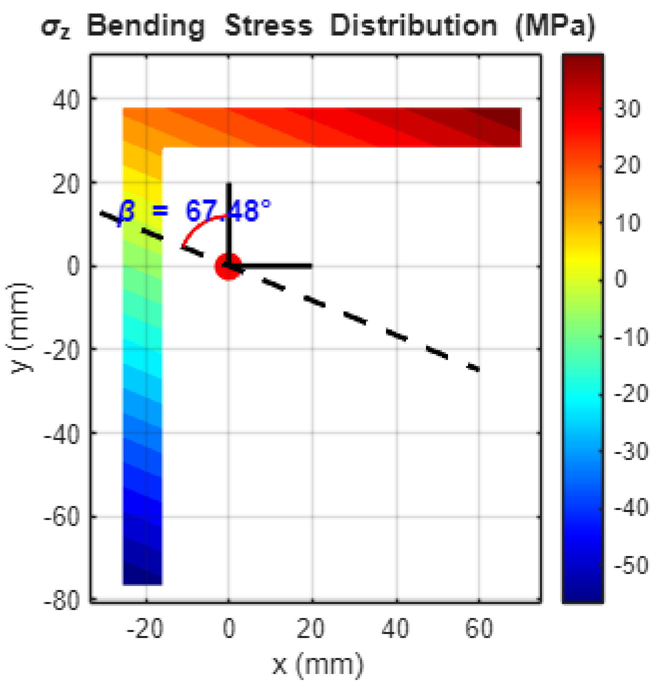
```
alpha =
22.5211
```

```
beta = 90-alpha
```

```
beta =
67.4789
```

```


% making a function for the stress function

sigma_z = @(x,y)(x_coeff*x + y_coeff*y);

[X, Y] = meshgrid(-40:0.5:70, -120:0.5:40);

% Masks for T-section geometry
inFlange = (X >= -x_cg) & (X <= 100-x_cg) & (Y >= y_cg-10) & (Y <= y_cg);
inWeb    = (X >= -x_cg) & (X <= 10-x_cg) & (Y >= -115+y_cg) & (Y <= -10+y_cg);
inSection = inFlange | inWeb;

Z = sigma_z(X, Y);
Z(~inSection) = NaN; % Hide points outside the section

% Plotting
figure;
contourf(X, Y, Z, 25, 'LineColor', 'none');
hold on;
colormap(jet); colorbar;
title('\sigma_z Bending Stress Distribution (MPa)');
xlabel('x (mm)'); ylabel('y (mm)');

% CG and Limited Axes
plot(0, 0, 'ro', 'MarkerSize', 10, 'MarkerFaceColor', 'r', 'DisplayName', 'CG');
plot([0, 0], [0, 20], 'k-', 'LineWidth', 2, 'DisplayName', 'Vertical Axis (x = 0)');
plot([0, 20], [0, 0], 'k-', 'LineWidth', 2, 'DisplayName', 'Horizontal Axis (y =
0)');

% Plot Neutral Axis
x_na = -60:1:60;
```

```
y_na = -tan(alpha * pi / 180) * x_na;
plot(x_na, y_na, 'k--', 'LineWidth', 2, 'DisplayName', 'Neutral Axis');

% =========================================================================
% NEW: ADD ANGLE ARC BETWEEN Y-AXIS AND NEUTRAL AXIS
% =========================================================================

% 1. Angles in polar coordinates (radians)
% Vertical axis (x = 0, y > 0) is at 90 degrees (pi/2)
theta_yAxis = pi / 2;

% Angle of the upper/positive-y segment of the Neutral Axis
theta_NA = atan2(-tan(alpha * pi / 180) * 1, 1);
if theta_NA < 0
    theta_NA = theta_NA + pi; % Ensure we take the branch in the upper half-plane
end

% 2. Generate arc points between the y-axis and Neutral Axis
r_arc = 12; % Radius of the arc (adjust size as needed)
t_arc = linspace(min(theta_yAxis, theta_NA), max(theta_yAxis, theta_NA), 100);
arc_x = r_arc * cos(t_arc);
arc_y = r_arc * sin(t_arc);

% Plot the arc line
plot(arc_x, arc_y, 'r-', 'LineWidth', 1.5, 'DisplayName', 'Angle \alpha');

% 3. Add text label at the midpoint of the arc
mid_theta = (theta_yAxis + theta_NA) / 2;
label_x = (r_arc + 3) * cos(mid_theta);
label_y = (r_arc + 3) * sin(mid_theta);

text(label_x, label_y, sprintf('\\beta = %.2f°', abs(90-alpha)), ...
    'FontSize', 11, 'FontWeight', 'bold', 'Color', 'b', ...
    'HorizontalAlignment', 'center');

% =========================================================================

axis equal;
grid on;

xlim([-33 75])
ylim([-81 51])
```




```
%legend('show');
sigma_z_A = sigma_z(100-x_cg,y_cg)
```

```
sigma_z_A =
45.0346
```

```
sigma_z_F = sigma_z(100-x_cg,y_cg-10)
```

```
sigma_z_F =
38.5214
```

```
sigma_z_D = sigma_z(-x_cg+10,y_cg-125)
```

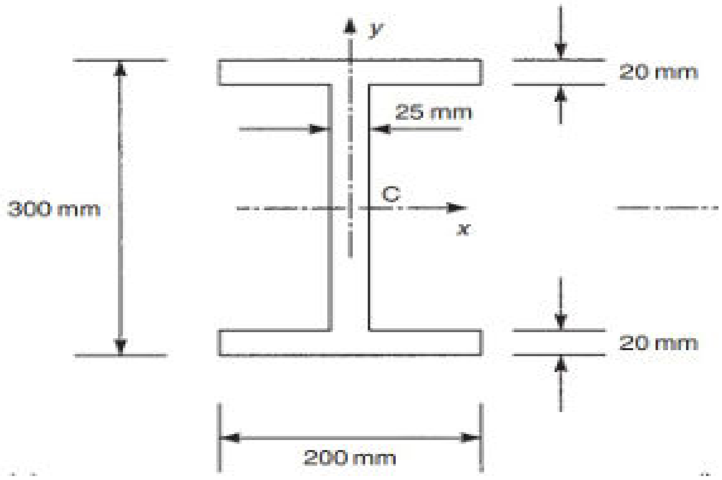
```
sigma_z_D =
-60.6875
```

```
sigma_z_C = sigma_z(-x_cg, y_cg - 125) % max
```

```
sigma_z_C =
-63.3882
```

Prob 3:

3. The beam section of Example 16.1 is subjected to a bending moment of 100 kN m applied in a plane parallel to the longitudinal axis of the beam but inclined at 30◦ to the left of vertical. The sense of the bending moment is clockwise when viewed from the left-hand edge of the beam section. Determine the distribution of direct stress.




```
%% Finding out the CG
% Area of each of the section in mm2
A1= 20*200; A2 = (300-40)*25;
x_cg = 100;  % from left
y_cg = 150;  % from top
x_bar= x_cg; y_bar=y_cg;  % according to the picture ( FROM LEFT AND TOP)
% floating-point with 2 decimals
fprintf('x_bar is %.4f and y_bar is %.4f\n', x_bar, y_bar);
```

```
x_bar is 100.0000 and y_bar is 150.0000
```

```
%%Finding Is in mm4
Ixx = 2*(200*20^3/12 + A1*140^2) + (25*(300-40)^3)/12
```

```
Ixx =
1.9368e+08
```

```
Iyy = 2*((20*200^3)/12 ) + ((300-40)*25^3)/12   % Calculate Iyy
```

```
Iyy =
2.7005e+07
```

```
Ixy = 0
```

```
Ixy =
0
```

```
% Moment acting on the plane 30 to the vertical plane (viewwd from leftside CW)
M = 100*1000*1000   ; % N mm at the
Mx = M*cosd(30); My = -M*sind(30);  % -ve sign is considered in the rest part
% Calculate the stress components
x_coeff = (My*Ixx - Mx*Ixy)/(Ixx*Iyy-Ixy^2)
```

```
x_coeff =
-1.8515
```

```
y_coeff = (Mx*Iyy - My*Ixy)/(Ixx*Iyy-Ixy^2)
```

```
y_coeff =
0.4471
```

```
% Calculating the angle of the of nutral axis
alpha = atan((My*Ixx - Mx*Ixy)/(Mx*Iyy - My*Ixy))*180/pi % in degrees
```

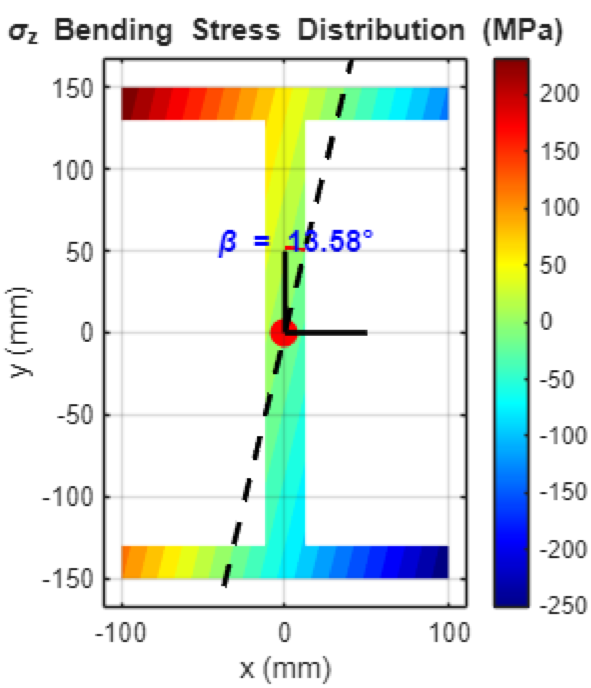
```
alpha =
-76.4231
```

```
beta=90-alpha
```

```
beta =
166.4231
```

```
% making a function for the stress function

sigma_z = @(x,y)(x_coeff*x + y_coeff*y);

[X, Y] = meshgrid(-110:0.51:110, -180:0.5:180);

% Masks for T-section geometry
inFlange1 = (X >= -x_cg) & (X <= x_cg) & (Y >= y_cg-20) & (Y <= y_cg);
inFlange2 = (X >= -x_cg) & (X <= x_cg) & (Y >= -y_cg) & (Y <= -y_cg+20);
inWeb    = (X >= -12.5) & (X <= 12.5) & (Y >= -y_cg+20) & (Y <= -20+y_cg);
inSection = inFlange1|inFlange2 | inWeb;

Z = sigma_z(X, Y);
Z(~inSection) = NaN; % Hide points outside the section

% Plotting
figure;
contourf(X, Y, Z, 25, 'LineColor', 'none');
hold on;
colormap(jet); colorbar;
title('\sigma_z Bending Stress Distribution (MPa)');
xlabel('x (mm)'); ylabel('y (mm)');

% CG and Limited Axes
plot(0, 0, 'ro', 'MarkerSize', 10, 'MarkerFaceColor', 'r', 'DisplayName', 'CG');
plot([0, 0], [0, 50], 'k-', 'LineWidth', 2, 'DisplayName', 'Vertical Axis (x = 0)');
plot([0, 50], [0, 0], 'k-', 'LineWidth', 2, 'DisplayName', 'Horizontal Axis (y =
0)');

% Plot Neutral Axis
x_na = -60:1:60;
y_na = -tan(alpha * pi / 180) * x_na;
plot(x_na, y_na, 'k--', 'LineWidth', 2, 'DisplayName', 'Neutral Axis');
```

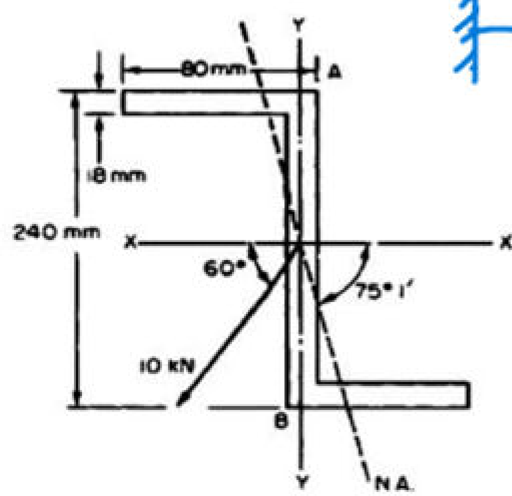
```matlab
% ==========================================================================
% NEW: ADD ANGLE ARC BETWEEN Y-AXIS AND NEUTRAL AXIS
% ==========================================================================

% 1. Angles in polar coordinates (radians)
% Vertical axis (x = 0, y > 0) is at 90 degrees (pi/2)
theta_yAxis = pi / 2;

% Angle of the upper/positive-y segment of the Neutral Axis
theta_NA = atan2(-tan(alpha * pi / 180) * 1, 1);
if theta_NA < 0
    theta_NA = theta_NA + pi; % Ensure we take the branch in the upper half-plane
end

% 2. Generate arc points between the y-axis and Neutral Axis
r_arc = 52; % Radius of the arc (adjust size as needed)
t_arc = linspace(min(theta_yAxis, theta_NA), max(theta_yAxis, theta_NA), 100);
arc_x = r_arc * cos(t_arc);
arc_y = r_arc * sin(t_arc);

% Plot the arc line
plot(arc_x, arc_y, 'r-', 'LineWidth', 1.5, 'DisplayName', 'Angle \alpha');

% 3. Add text label at the midpoint of the arc
mid_theta = (theta_yAxis + theta_NA) / 2;
label_x = (r_arc + 3) * cos(mid_theta);
label_y = (r_arc + 3) * sin(mid_theta);

text(label_x, label_y, sprintf('\\beta = %.2f°', abs(90+alpha)), ...
    'FontSize', 11, 'FontWeight', 'bold', 'Color', 'b', ...
    'HorizontalAlignment', 'center');

% ==========================================================================

axis equal;
grid on;

xlim([-110 110])
ylim([-168 168])
```

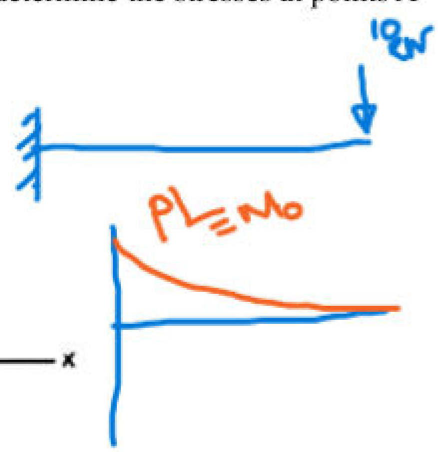


Prob 4:

4. A horizontal cantilever 2 m long is constructed from the Z-section shown below. A load of 10 KN is applied to the end of the cantilever at an angle of 60°to the horizontal as shown. Assuming that no twisting moment is applied to the section, determine the stresses at points A and B.

(Ixx = 48.3 x 10−6m⁴, Iyy = 4.4 x 10−6m⁴)






```
%% Finding out the CG
% Area of each of the section in mm2
A1= 8*80; A2 = (240-16)*8;
x_cg = 80-4;  % from left
y_cg = 120;  % from top
x_bar= x_cg; y_bar=y_cg;  % according to the picture ( FROM LEFT AND TOP)
```

```
% floating-point with 2 decimals
fprintf('x_bar is %.4f and y_bar is %.4f\n', x_bar, y_bar);
```

```
x_bar is 76.0000 and y_bar is 120.0000
```

```
% Calculate the second moments of area for the new sections
Ixx = 2*((80*8^3)/12 + A1*(y_cg-4)^2) + (8*(240-16)^3)/12
```

```
Ixx =
24723456
```

```
Iyy = 2*((8*80^3)/12 + A1*(x_cg-40)^2) + ((240-16)*8^3)/12
```

```
Iyy =
2351104
```

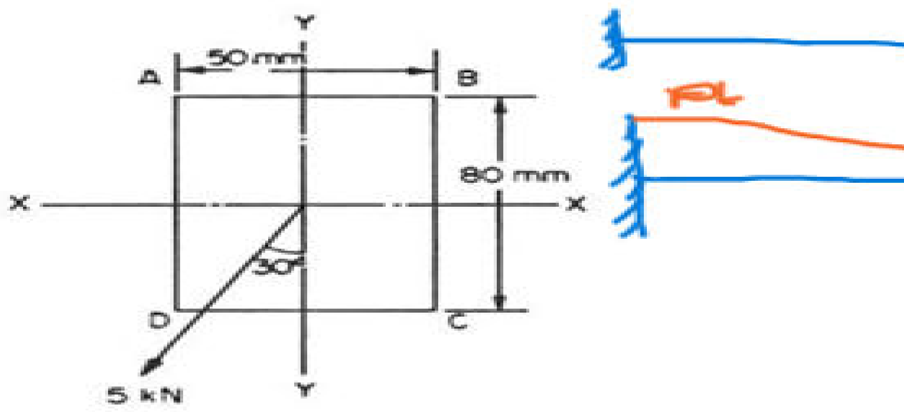
```
Ixy = A1*(x_cg-40)*(-y_cg+4) + A1*(-x_cg+40)*(y_cg-4)
```

```
Ixy =
-5345280
```

```
%% Current Ixx and Iyy , Ixy Are not matching with the given details
```

Prob 5:

5. A rectangular section beam 80 mm x 50 mm is arranged as a cantilever
loaded at its free end with a load of 5 kN inclined at an angle of 30° to the
in Fig. 2. Determine the position and magnitude of the greatest tensile str




```
%% Finding out the CG
% Area of each of the section in mm2
A1= 50*80 ;
x_cg = 25;  % from left
y_cg = 40;  % from top
x_bar= x_cg; y_bar=y_cg;  % according to the picture ( FROM LEFT AND TOP)
% floating-point with 2 decimals
fprintf('x_bar is %.4f and y_bar is %.4f\n', x_bar, y_bar);
```

```
x_bar is 25.0000 and y_bar is 40.0000
```

```
% CAlculating the Is
Ixx = 50*80^3/12
```

```
Ixx =
2.1333e+06
```

```
Iyy= 80*50^3/12
```

```
Iyy =
8.3333e+05
```

```
Ixy = 0
```

```
Ixy =
0
```

```
% moment calc
M = 5000*1.3*1000;  % acting on the plane at angle 30 with the -ve y axis in
clockwise direction
% So net moment will be in 60deg in CCW from -ve y axis
Mx = M*sind(60); My=M*cosd(60);
% Calculate the stress components
x_coeff = (My*Ixx - Mx*Ixy)/(Ixx*Iyy-Ixy^2)
```

```
x_coeff =
3.9000
```

```
y_coeff = (Mx*Iyy - My*Ixy)/(Ixx*Iyy-Ixy^2)
```

```
y_coeff =
2.6387
```

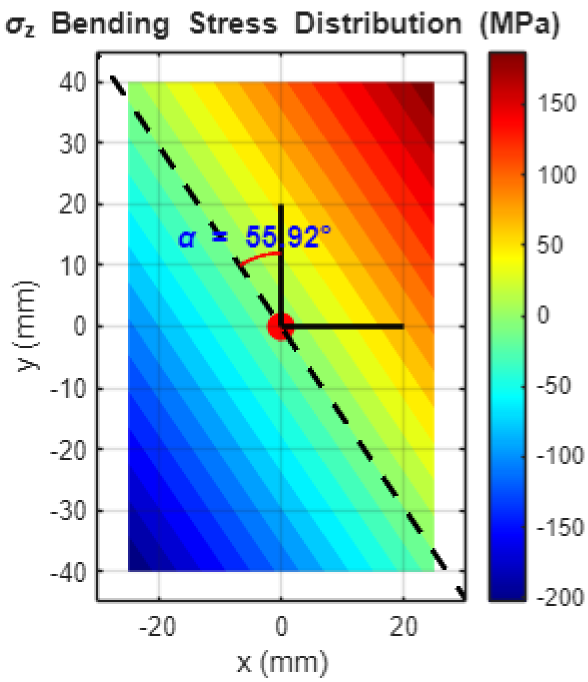
```
% Calculating the angle of the of nutral axis
alpha = atan((My*Ixx - Mx*Ixy)/(Mx*Iyy - My*Ixy))*180/pi ;% in degrees
beta = alpha
```

```
beta =
55.9184
```

```
% making a function for the stress function

sigma_z = @(x,y)(x_coeff*x + y_coeff*y);

[X, Y] = meshgrid(-30:0.5:30, -50:0.5:50);

% Masks for T-section geometry
inFlange = (X >= -x_cg) & (X <= x_cg) & (Y >= -y_cg) & (Y <= y_cg);

inSection = inFlange;

Z = sigma_z(X, Y);
Z(~inSection) = NaN; % Hide points outside the section

% Plotting
figure;
```

```
contourf(X, Y, Z, 25, 'LineColor', 'none');
hold on;
colormap(jet); colorbar;
title('\sigma_z Bending Stress Distribution (MPa)');
xlabel('x (mm)'); ylabel('y (mm)');

% CG and Limited Axes
plot(0, 0, 'ro', 'MarkerSize', 10, 'MarkerFaceColor', 'r', 'DisplayName', 'CG');
plot([0, 0], [0, 20], 'k-', 'LineWidth', 2, 'DisplayName', 'Vertical Axis (x = 0)');
plot([0, 20], [0, 0], 'k-', 'LineWidth', 2, 'DisplayName', 'Horizontal Axis (y =
0)');

% Plot Neutral Axis
x_na = -60:1:60;
y_na = -tan(alpha * pi / 180) * x_na;
plot(x_na, y_na, 'k--', 'LineWidth', 2, 'DisplayName', 'Neutral Axis');

% ========================================================================
% NEW: ADD ANGLE ARC BETWEEN Y-AXIS AND NEUTRAL AXIS
% ========================================================================

% 1. Angles in polar coordinates (radians)
% Vertical axis (x = 0, y > 0) is at 90 degrees (pi/2)
theta_yAxis = pi / 2;

% Angle of the upper/positive-y segment of the Neutral Axis
theta_NA = atan2(-tan(alpha * pi / 180) * 1, 1);
if theta_NA < 0
    theta_NA = theta_NA + pi; % Ensure we take the branch in the upper half-plane
end

% 2. Generate arc points between the y-axis and Neutral Axis
r_arc = 12; % Radius of the arc (adjust size as needed)
t_arc = linspace(min(theta_yAxis, theta_NA), max(theta_yAxis, theta_NA), 100);
arc_x = r_arc * cos(t_arc);
arc_y = r_arc * sin(t_arc);

% Plot the arc line
plot(arc_x, arc_y, 'r-', 'LineWidth', 1.5, 'DisplayName', 'Angle \alpha');

% 3. Add text label at the midpoint of the arc
mid_theta = (theta_yAxis + theta_NA) / 2;
label_x = (r_arc + 3) * cos(mid_theta);
label_y = (r_arc + 3) * sin(mid_theta);

text(label_x, label_y, sprintf('\\alpha = %.2f°', abs(alpha)), ...
    'FontSize', 11, 'FontWeight', 'bold', 'Color', 'b', ...
    'HorizontalAlignment', 'center');

% ========================================================================
```

```
axis equal;
grid on;
%legend('show');

xlim([-30 30])
ylim([-45 45])
```




```
% FInding some stress points (MPa)
sigma_z_B = sigma_z(x_cg,y_cg)
```

```
sigma_z_B =
203.0468
```

```
sigma_z_D = sigma_z(-x_cg,-y_cg)
```

```
sigma_z_D =
-203.0468
```

Prob 6 :

6. A cantilever of length 1.2 m and of the cross section shown in Fig. 3 carries a ver
of 10 kN at its outer end, the line of action being parallel with the longer leg and arı
pass through the shear centre of the section (i.e. there is no twisting of the section).
from first principles, find the stress set up in the section at points A, B and C, given
centroid is located as shown. Also determine the angle of inclination of the N.A.

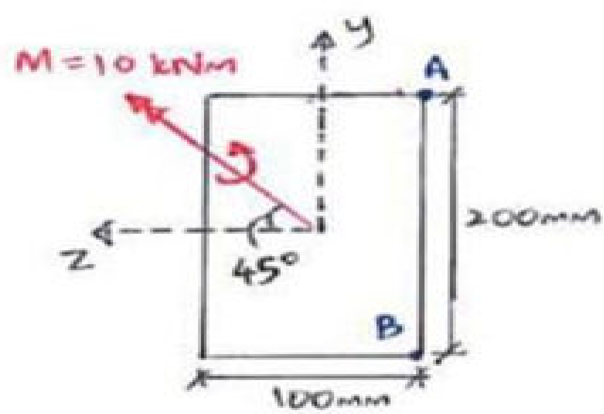
($I_{xx} = 4 \times 10^{-6}\ m^4$, $I_{yy} = 1.08 \times 10^{-6}\ m^4$) P= 10kN

7.

```
%finding CG
%%% Diagram of the Question is not given
```

Problem 7:

7.



Calculate the bending stress at points **A** and B.

```
%% Finding out the CG
% Area of each of the section in mm2
A1= 50*80 ;
z_cg = 50;  % from left
y_cg = 100;  % from top
z_bar= z_cg; y_bar=y_cg;  % according to the picture ( FROM LEFT AND TOP)
% floating-point with 2 decimals
```

```
fprintf('x_bar is %.4f and y_bar is %.4f\n', z_bar, y_bar);
```

```
x_bar is 50.0000 and y_bar is 100.0000
```

```


% CAlculating the Is
Izz = 100*200^3/12
```

```
Izz =
6.6667e+07
```

```
Iyy= 200*100^3/12
```

```
Iyy =
1.6667e+07
```

```
Iyz = 0
```

```
Iyz =
0
```

```
% moment calc (N_mm)
M = 10000*1000;  % acting on the plane at angle 30 with the -ve y axis in clockwise
direction
% So net moment will be in 60deg in CCW from -ve y axis
Mz = M*sind(45); My=M*cosd(45);
% Calculate the stress components
y_coeff = -(Mz*Iyy + My*Iyz)/(-Izz*Iyy + Iyz^2)
```

```
y_coeff =
0.1061
```

```
z_coeff = (-Mz*Iyz - My*Izz)/(-Izz*Iyy + Iyz^2)
```

```
z_coeff =
0.4243
```

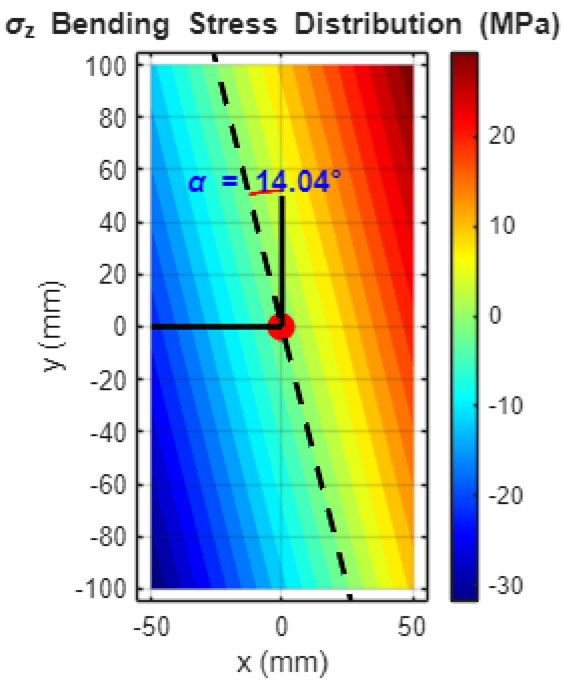
```
% Calculating the angle of the of nutral axis
alpha = atan((Mz*Iyy + My*Iyz)/(Mz*Iyz + My*Izz))*180/pi ;% in degrees
beta = alpha
```

```
beta =
14.0362
```

```
% making a function for the stress function

sigma_z = @(z,y)(z_coeff*z + y_coeff*y);

[Z, Y] = meshgrid(-55:0.5: 55, -105:0.5:105);

% Masks for T-section geometry
inFlange = (Z >= -z_cg) & (Z <= z_cg) & (Y >= -y_cg) & (Y <= y_cg);

inSection = inFlange;

X = sigma_z(Z, Y);
```

```
X(~inSection) = NaN; % Hide points outside the section

% Plotting
figure;
contourf(Z, Y, X, 25, 'LineColor', 'none');
hold on;
colormap(jet); colorbar;
title('\sigma_z Bending Stress Distribution (MPa)');
xlabel('x (mm)'); ylabel('y (mm)');

% CG and Limited Axes
plot(0, 0, 'ro', 'MarkerSize', 10, 'MarkerFaceColor', 'r', 'DisplayName', 'CG');
plot([0, 0], [0, 50], 'k-', 'LineWidth', 2, 'DisplayName', 'Vertical Axis (z = 0)');
plot([0, -50], [0, 0], 'k-', 'LineWidth', 2, 'DisplayName', 'Horizontal Axis (y =
0)');

% Plot Neutral Axis
z_na = -60:1:60;
y_na = -tan((pi/2)-alpha * pi / 180) * z_na;
plot(z_na, y_na, 'k--', 'LineWidth', 2, 'DisplayName', 'Neutral Axis');

% ========================================================================
% NEW: ADD ANGLE ARC BETWEEN Y-AXIS AND NEUTRAL AXIS
% ========================================================================

% 1. Angles in polar coordinates (radians)
% Vertical axis (x = 0, y > 0) is at 90 degrees (pi/2)
theta_yAxis = pi / 2;

% Angle of the upper/positive-y segment of the Neutral Axis
theta_NA = atan2(-tan((pi/2) -alpha * pi / 180) * 1, 1);
if theta_NA < 0
    theta_NA = theta_NA + pi; % Ensure we take the branch in the upper half-plane
end

% 2. Generate arc points between the y-axis and Neutral Axis
r_arc = 52; % Radius of the arc (adjust size as needed)
t_arc = linspace(min(theta_yAxis, theta_NA), max(theta_yAxis, theta_NA), 100);
arc_x = r_arc * cos(t_arc);
arc_y = r_arc * sin(t_arc);

% Plot the arc line
plot(arc_x, arc_y, 'r-', 'LineWidth', 1.5, 'DisplayName', 'Angle \alpha');

% 3. Add text label at the midpoint of the arc
mid_theta = (theta_yAxis + theta_NA) / 2;
label_x = (r_arc + 3) * cos(mid_theta);
label_y = (r_arc + 3) * sin(mid_theta);

text(label_x, label_y, sprintf('\\alpha = %.2f°', abs(alpha)), ...
```

```
    'FontSize', 11, 'FontWeight', 'bold', 'Color', 'b', ...
    'HorizontalAlignment', 'center');

% ========================================================================

axis equal;
grid on;
%legend('show');

xlim([-55 55])
ylim([-105 105])
```




```
% FInding some stress points (MPa)
sigma_x_A = sigma_z(z_cg,y_cg)
```

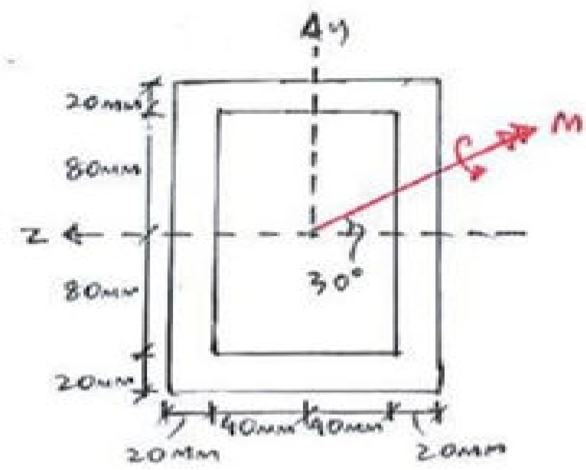
```
sigma_x_A =
31.8198
```

```
sigma_x_B = sigma_z(z_cg,-y_cg) % Max
```

```
sigma_x_B =
10.6066
```

Problem 8:

8.



If $\sigma_{b,max} = 160$ **MPa**, calculate the maximum moment **M** that can be applied. Als
orientation of the neutral-axis (NA).

```
%% Finding out the CG
% Area of each of the section in mm2
A1= 200*120 ; A2= -160*80;
z_cg = 60*(A1 + A2)/(A1+A2);  % from left
y_cg = (A1*100 + A2*100)/(A1+A2);  % from top
z_bar= z_cg; y_bar=y_cg;  % according to the picture ( FROM LEFT AND TOP)
% floating-point with 2 decimals
fprintf('z_bar is %.4f and y_bar is %.4f\n', z_bar, y_bar);
```

```
z_bar is 60.0000 and y_bar is 100.0000
```

```
% CAlculating the Is
Izz = 120*200^3/12 - 80*160^3/12
```

```
Izz =
5.2693e+07
```

```
Iyy= 120^3*200^1/12 - 80^3*160^1/12
```

```
Iyy =
2.1973e+07
```

```
Iyz = 0
```

```
Iyz =
0
```

```
% moment calc (N_mm)
```

```
M = 10000*1000;  % acting on the plane at angle 30 with the -ve y axis in clockwise
direction
% So net moment will be in 60deg in CCW from -ve y axis
Mz = M*sind(45); My=M*cosd(45);
% Calculate the stress components
y_coeff = -(Mz*Iyy + My*Iyz)/(-Izz*Iyy + Iyz^2)
```

```
y_coeff =
0.1342
```

```
z_coeff = (-Mz*Iyz - My*Izz)/(-Izz*Iyy + Iyz^2)
```

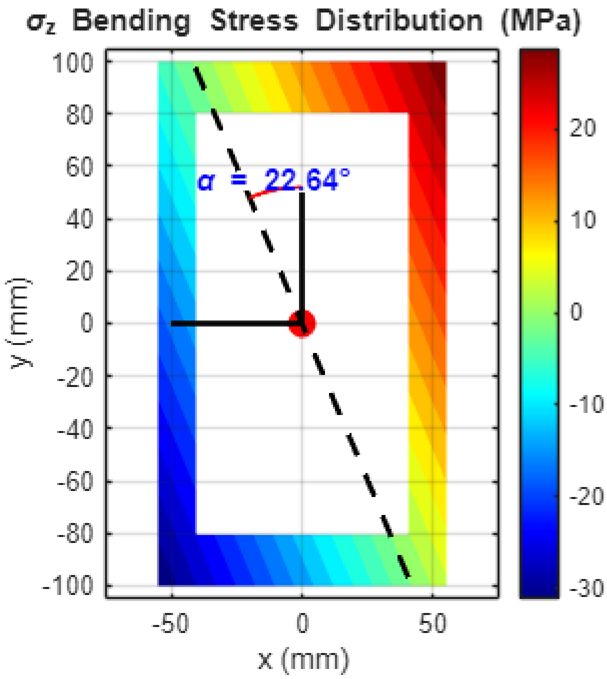
```
z_coeff =
0.3218
```

```
% Calculating the angle of the of nutral axis
alpha = atan((Mz*Iyy + My*Iyz)/(Mz*Iyz + My*Izz))*180/pi % in degrees
```

```
alpha =
22.6363
```

```
% making a function for the stress function

sigma_z = @(z,y)(z_coeff*z + y_coeff*y);

[Z, Y] = meshgrid(-55:0.5: 55, -105:0.5:105);

% Masks for T-section geometry
inFlange = (Z >= -z_cg) & (Z <= z_cg) & (Y >= -y_cg) & (Y <= y_cg);
outFlange = (Z >= -(z_cg-20)) & (Z <= (z_cg-20)) & (Y >= -(y_cg-20)) & (Y <=
(y_cg-20));
inSection = inFlange & ~outFlange ;

X = sigma_z(Z, Y);
X(~inSection) = NaN; % Hide points outside the section

% Plotting
figure;
contourf(Z, Y, X, 25, 'LineColor', 'none');
hold on;
colormap(jet); colorbar;
title('\sigma_z Bending Stress Distribution (MPa)');
xlabel('x (mm)'); ylabel('y (mm)');

% CG and Limited Axes
plot(0, 0, 'ro', 'MarkerSize', 10, 'MarkerFaceColor', 'r', 'DisplayName', 'CG');
plot([0, 0], [0, 50], 'k-', 'LineWidth', 2, 'DisplayName', 'Vertical Axis (z = 0)');
plot([0, -50], [0, 0], 'k-', 'LineWidth', 2, 'DisplayName', 'Horizontal Axis (y =
0)');

% Plot Neutral Axis
z_na = -60:1:60;
y_na = -tan((pi/2)-alpha * pi / 180) * z_na;
```

```
plot(z_na, y_na, 'k--', 'LineWidth', 2, 'DisplayName', 'Neutral Axis');

% ===========================================================================
% NEW: ADD ANGLE ARC BETWEEN Y-AXIS AND NEUTRAL AXIS
% ===========================================================================

% 1. Angles in polar coordinates (radians)
% Vertical axis (x = 0, y > 0) is at 90 degrees (pi/2)
theta_yAxis = pi / 2;

% Angle of the upper/positive-y segment of the Neutral Axis
theta_NA = atan2(-tan((pi/2) -alpha * pi / 180) * 1, 1);
if theta_NA < 0
    theta_NA = theta_NA + pi; % Ensure we take the branch in the upper half-plane
end

% 2. Generate arc points between the y-axis and Neutral Axis
r_arc = 52; % Radius of the arc (adjust size as needed)
t_arc = linspace(min(theta_yAxis, theta_NA), max(theta_yAxis, theta_NA), 100);
arc_x = r_arc * cos(t_arc);
arc_y = r_arc * sin(t_arc);

% Plot the arc line
plot(arc_x, arc_y, 'r-', 'LineWidth', 1.5, 'DisplayName', 'Angle \alpha');

% Add text label at the midpoint of the arc
mid_theta = (theta_yAxis + theta_NA) / 2;
label_x = (r_arc + 3) * cos(mid_theta);
label_y = (r_arc + 3) * sin(mid_theta);

text(label_x, label_y, sprintf('\\alpha = %.2f°', abs(alpha)), ...
    'FontSize', 11, 'FontWeight', 'bold', 'Color', 'b', ...
    'HorizontalAlignment', 'center');

% ===========================================================================

% 1. Set equal axis aspect ratio (1 unit X = 1 unit Y visually)
axis equal;
grid on;
%legend('show');

% 2. Match the axis limits range (210 units total on both X and Y)
xlim([-75 75]);
ylim([-105 105]);

% 3. Force the physical plotting box to be a square
gca1 = gca;
gca1.PlotBoxAspectRatio = [1 1 1];
```



σ_z Bending Stress Distribution (MPa)
α = 22.64°
y (mm)
x (mm)
100
80
60
40
20
0
-20
-40
-60
-80
-100
-50
0
50
20
10
0
-10
-20
-30

Solution :

Tut-1 8: $M_z = -M\sqrt{3}/2$ N.m

$M_y = M/2$ N.m

Now, $\sigma_b = \left(-\frac{M_z \cdot y}{I_{z-z}} + \frac{M_y \cdot z}{I_{yy}}\right) \times 10^3$ MPa.

Now, $I_{zz} = \frac{120 \times 200^3}{12} - \frac{80 \times 160^3}{12}$ mm^4

$I_{yy} = \frac{120^3 \times 200}{12} - \frac{80^3 \times 160}{12}$ mm^4 ; $\boxed{I_{yy} = 2.1}$

$\boxed{I_{zz} = 5.269 \times 10^7 \text{ mm}^4}$

$\therefore \sigma_b = \frac{M}{2}\left(\frac{\sqrt{3}\, y}{I_{zz}} + \frac{z}{I_{yy}}\right) \times 10^3$ MPa

Now, $\sigma_b|_{max}$ at $y = 100$ mm & $z =$

So, $\sigma_b|_{max} = \frac{M}{2} \times 10^3 \left(\frac{\sqrt{3}(100)}{I_{zz}} + \frac{60}{I_{yy}}\right)$

$$\therefore M = \frac{160 \times 2 \times 10^{-3}}{\frac{\sqrt{3} \times 100}{5.269 \times 10^7} + \frac{60}{2.197 \times 10^7}}$$

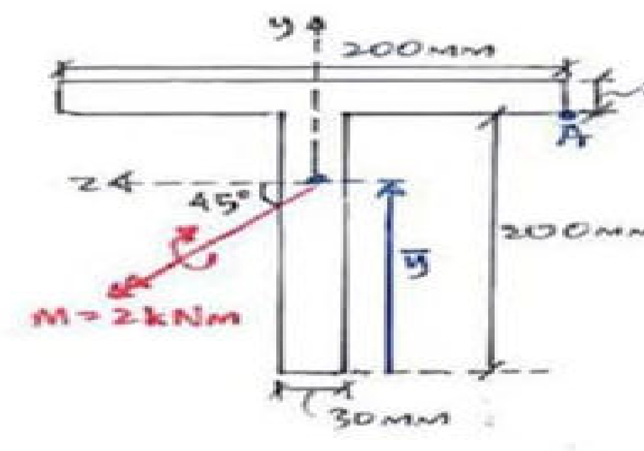
$M = 53171.65$ N.m $\Rightarrow \boxed{M =}$

Orientation of N.A.:—

```
```

PROBLEM 9:

9.



Find the bending stress developed at point A.

```
%% Finding out the CG
% Area of each of the section in mm2
A1= 200*30 ; A2= 200*30;
z_cg = (A1*100 + A2*100)/(A1+A2);  % from left
y_cg = (A1*15 + A2*(30+100))/(A1+A2);  % from top
z_bar= z_cg; y_bar=230-y_cg;  % according to the picture ( FROM LEFT AND TOP)
% floating-point with 2 decimals
fprintf('z_bar is %.4f and y_bar is %.4f\n', z_bar, y_bar);
```

```
z_bar is 100.0000 and y_bar is 157.5000
```

```

% CAlculating the Is
Izz = 200*30^3/12 + A1*(15-y_cg)^2 +  30*200^3/12 +A2*(30+100-y_cg)^2
```

```
Izz =
60125000
```

```
Iyy= 200^3*30^1/12 + 30^3*200^1/12
```

```
Iyy =
20450000
```

```
Iyz = 0
```

```
Iyz =
0
```

```
% moment calc (N_mm)
M = 2000*1000;  % acting on the plane at angle 30 with the -ve y axis in clockwise
direction
% So net moment will be in 60deg in CCW from -ve y axis
My = -M*sind(45); Mz=M*cosd(45);
% Calculate the stress components
y_coeff = -(Mz*Iyy + My*Iyz)/(-Izz*Iyy + Iyz^2)
```

```
y_coeff =
0.0235
```

```
z_coeff = (-Mz*Iyz - My*Izz)/(-Izz*Iyy + Iyz^2)
```

```
z_coeff =
-0.0692
```

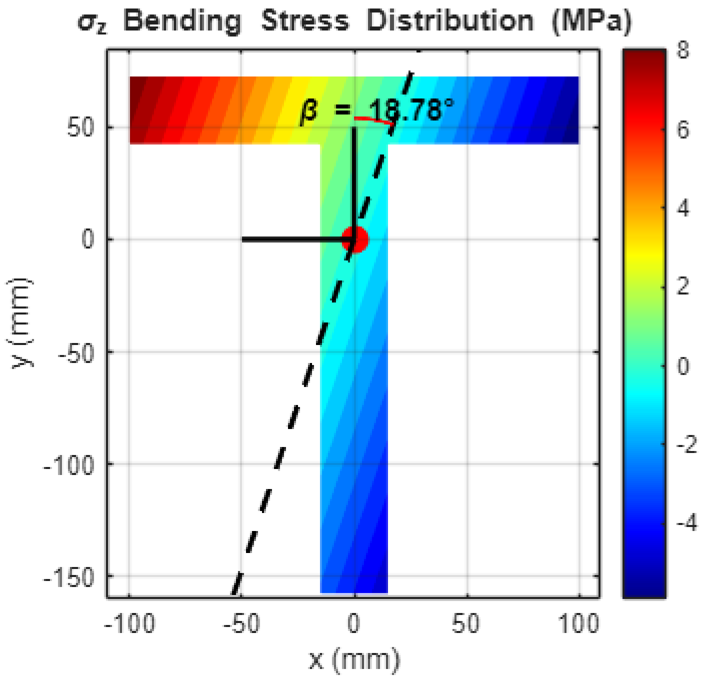
```
% Calculating the angle of the of nutral axis
alpha = atan((Mz*Iyy + My*Iyz)/(Mz*Iyz + My*Izz))*180/pi; % in degrees
beta = 2*90+alpha
```

```
beta =
161.2156
```

```
% making a function for the stress function

sigma_z = @(z,y)(z_coeff*z + y_coeff*y);

[Z, Y] = meshgrid(-110:0.5: 110, -160:0.5:80);

% Masks for T-section geometry
inFlange = (Z >= -z_cg) & (Z <= z_cg) & (Y >= (y_cg-30)) & (Y <= y_cg);
WEDGE = (Z >= -(15)) & (Z <= (15)) & (Y >= -(y_bar)) & (Y <= (200-y_bar));
inSection = inFlange |WEDGE  ;

X = sigma_z(Z, Y);
X(~inSection) = NaN; % Hide points outside the section

% Plotting
figure;
contourf(Z, Y, X, 25, 'LineColor', 'none');
hold on;
colormap(jet); colorbar;
title('\sigma_z Bending Stress Distribution (MPa)');
xlabel('x (mm)'); ylabel('y (mm)');

% CG and Limited Axes
plot(0, 0, 'ro', 'MarkerSize', 10, 'MarkerFaceColor', 'r', 'DisplayName', 'CG');
plot([0, 0], [0, 50], 'k-', 'LineWidth', 2, 'DisplayName', 'Vertical Axis (z = 0)');
plot([0, -50], [0, 0], 'k-', 'LineWidth', 2, 'DisplayName', 'Horizontal Axis (y =
0)');
```

```
% Plot Neutral Axis
z_na = -60:1:60;
y_na = -tan((pi/2)-alpha * pi / 180) * z_na;
plot(z_na, y_na, 'k--', 'LineWidth', 2, 'DisplayName', 'Neutral Axis');

% ==========================================================================
% NEW: ADD ANGLE ARC BETWEEN Y-AXIS AND NEUTRAL AXIS
% ==========================================================================

% 1. Angles in polar coordinates (radians)
% Vertical axis (x = 0, y > 0) is at 90 degrees (pi/2)
theta_yAxis = pi / 2;

% Angle of the upper/positive-y segment of the Neutral Axis
theta_NA = atan2(-tan((pi/2) -alpha * pi / 180) * 1, 1);
if theta_NA < 0
    theta_NA = theta_NA + pi; % Ensure we take the branch in the upper half-plane
end

% 2. Generate arc points between the y-axis and Neutral Axis
r_arc = 54; % Radius of the arc (adjust size as needed)
t_arc = linspace(min(theta_yAxis, theta_NA), max(theta_yAxis, theta_NA), 100);
arc_x = r_arc * cos(t_arc);
arc_y = r_arc * sin(t_arc);

% Plot the arc line
plot(arc_x, arc_y, 'r-', 'LineWidth', 1.5, 'DisplayName', 'Angle \alpha');

% Add text label at the midpoint of the arc
mid_theta = (theta_yAxis + theta_NA) / 2;
label_x = (r_arc + 3) * cos(mid_theta);
label_y = (r_arc + 3) * sin(mid_theta);

text(label_x, label_y, sprintf('\\beta = %.2f°', abs(alpha)), ...
    'FontSize', 11, 'FontWeight', 'bold', 'Color', 'k', ...
    'HorizontalAlignment', 'center');

% ==========================================================================

% 1. Set equal axis aspect ratio (1 unit X = 1 unit Y visually)
axis equal;
grid on;
%legend('show');

% 2. Match the axis limits range (210 units total on both X and Y)
xlim([-75 75]);
ylim([-235 235]);

% 3. Force the physical plotting box to be a square
```

```
gca1 = gca;
gca1.PlotBoxAspectRatio = [1 1 1];

xlim([-110 109])
ylim([-160 85])
```




```
% FInding some stress points (MPa)
sigma_x_A = sigma_z(z_cg,y_cg-30)
```

```
sigma_x_A =
-5.9158
```